

Nirma University

Institute of Technology

Semester End Examination (IR), May - 2026

M. Tech. in Computer Science and Engineering /

M. Tech. in Computer Science and Engineering (Data Science), Semester-II

6CS272ME25 Deep Learning and Applications

Roll/Exam No

25MCOO05

Supervisor's

Initial with date

S.A. 08-05-26

Time: 3 Hours.

Max. Marks: 100

Instructions: 1. All questions are compulsory. (No optional Questions)

2. Use section-wise separate answer books

3. Figures to the right indicate full marks

4. Assume suitable data if required.

5. Use of a scientific calculator is permitted.

SECTION-I

- Q.1 Answer the following questions: [16]**
- (a) CLO1 Explain Triplet Loss? What applications will need this type of loss function? Define the Siamese network architecture and its working during training and during inference. [06]
BL:2
- (b) CLO1 Draw a block diagram to translate from Gujarati to English. The vocab size for Gujarati is 15k, and the vocab size for English is 10K. The embedding size used is 100 in both languages. The hidden state is represented by a vector of size 500. Draw the block diagram, and calculate the number of learnable parameters. [10]
BL:3
- Q.2 Answer the following questions: [18]**
- (a) CLO1 What is gradient and back propagation in time for the sequence models? Explain Backprop in Sequence models. [06]
BL:2
- (b) CLO3 IMDB Data for sentiment classification is to be done. Data contains paragraphs that define the movie and the paragraph's sentiment across five classes. Give the steps for developing a model to classify a new comment. [12]
BL:5
- Choice of the model
 - Preparing the dataset
 - Training the model and using the model
 - Block diagram of the model
- Q.3 Answer the following questions: [16]**
- (a) CLO3 What is the mean average precision(MAP) in the context of Object detection? What is non-max suppression in the Yolo Model? Explain in detail the functioning of the Yolo model and the performance metric to evaluate them. [10]
BL:3
- (b) CLO1 The Inception model uses inception blocks. To minimize the number of learnable parameters, it uses the concept of factorization. Explain how it reduces the number of [06]
BL:4

learnable parameters while serving the same functionality. Take an appropriate example to explain.

SECTION-II

- Q.4** **Answer the following questions.** [18]
- (a) CLO2 Segmentation uses transpose convolution and also Atrous [08]
BL:3 convolution. Explain the difference between the two using an appropriate example.
- (b) CLO2 Consider a simplified version of a convolutional neural [10]
BL:2 network inspired by VGGNet.

You are given the following 4×4 grayscale input image X and a 3×3 filter K

$$X = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 3 & 1 & 2 & 2 \\ 0 & 1 & 3 & 1 \\ 2 & 2 & 1 & 0 \end{bmatrix} \quad K = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Perform :

1. **Convolution Operation**

Apply the filter to the input image with:

- Stride = 1
- No padding

Compute the resulting feature map.

2. **Activation Function**

Apply the **ReLU** activation function to the feature map obtained in part (1).

3. **Pooling Layer**

Apply **2×2 max pooling** to the result from part (2).

4. **Fully Connected Layer**

Flatten the pooled output and compute the final output using:

- Weight = 2
- Bias = 1

Q.5 **Answer the following questions.** [16]

- (a) CLO2 Give 3 real-life problems that can be solved by [10]
BL:4 reinforcement learning, with proper explanation.
- (b) CLO1 What is another name for pixel-level classification? What [06]
BL:4 is its explanation? Explain any one commonly used model used for pixel-level classification.

Q.6 **Answer the following questions.** [16]

- (a) CLO3 In GANs, we do not have an input-output pair dataset. [10]
BL:6 The cost function is typically the gap between the expected and generated outputs. Explain the cost function in GAN networks.
- (b) CLO2 Explain the concept of Transfer learning? What are the [06]
BL:3 different scenarios where transfer learning can be used? In which scenario should transfer learning not be used at all.